Skill & Subskills-2

61. Deep Learning

Preparedness target: STRONG

This is a core ML interview area. You should understand the mechanics rather than merely knowing how to call PyTorch APIs.

Subskills

1. Neural-network fundamentals
   - neurons
   - layers
   - weights
   - biases
   - forward propagation
2. Activation functions
   - sigmoid
   - tanh
   - ReLU
   - Leaky ReLU
   - softmax
   - activation-function selection
3. Loss functions
   - MSE
   - binary cross-entropy
   - categorical cross-entropy
   - loss interpretation

4. Backpropagation
    - computation graph
    - chain rule
    - gradients
    - backward propagation
5. Gradient descent
    - learning rate
    - batch gradient descent
    - stochastic gradient descent
    - mini-batch gradient descent
6. Weight initialization
    - random initialization
    - Xavier/Glorot
    - He initialization
    - initialization problems
7. Vanishing gradients
    - causes
    - symptoms
    - mitigation
8. Exploding gradients
    - causes
    - symptoms
    - gradient clipping
9. Optimization algorithms
    - SGD
    - momentum

- RMSProp
- Adam
- optimizer trade-offs

10. Learning-rate strategies
    - fixed learning rate
    - decay
    - schedulers
    - warmup
    - adaptive learning rates
11. Overfitting
    - training vs validation performance
    - regularization
    - dropout
    - early stopping
12. Batch normalization
    - normalization
    - training behavior
    - inference behavior
    - benefits
13. Dropout
    - random neuron dropping
    - training vs inference
    - regularization effect
14. Weight decay
    - L2 regularization
    - parameter shrinkage

- optimizer interaction

15. Training/validation curves
    - loss curves
    - accuracy curves
    - diagnosing underfitting
    - diagnosing overfitting
16. Batch size
    - small vs large batches
    - gradient noise
    - memory requirements
    - training speed
17. Epochs and iterations
    - epoch
    - batch
    - iteration
    - training schedule
18. Hyperparameter tuning
    - learning rate
    - batch size
    - architecture
    - regularization
    - optimizer
19. Transfer learning
    - pretrained models
    - feature extraction
    - fine-tuning

20. GPU training
    - GPU acceleration
    - tensor devices
    - memory usage
    - batching
21. Model capacity
    - parameters
    - depth
    - width
    - capacity vs generalization
22. Neural-network debugging
    - NaN loss
    - exploding gradients
    - incorrect labels
    - shape mismatch
    - dead neurons
23. Deep-learning evaluation
    - validation
    - test set
    - task-specific metrics
    - error analysis
24. Deep-learning engineering
    - checkpoints
    - reproducibility
    - experiment tracking
    - inference pipeline

62. Convolutional Neural Networks (CNNs)

Preparedness target: STRONG

Especially relevant to your genomics/sequence work and computer-vision-style ML interviews.

Subskills

1. Convolution intuition
    - kernels
    - receptive fields
    - local patterns
    - parameter sharing
2. 1D convolution
    - sequence convolution
    - kernel over sequence
    - genomic sequence applications
3. 2D convolution
    - image representation
    - spatial kernels
    - feature maps
4. Convolution parameters
    - kernel size
    - stride
    - padding
    - number of channels
5. Output-size calculation

- input dimensions
- kernel
- stride
- padding

6. Feature maps
   - channels
   - learned features
   - hierarchical representations
7. Pooling
   - max pooling
   - average pooling
   - downsampling
8. Receptive field
   - local context
   - stacked convolutions
   - receptive-field growth
9. CNN architecture
   - convolution blocks
   - activation
   - normalization
   - pooling
   - classification head
10. CNN parameter count
    - kernel parameters
    - channels
    - bias

- comparing dense vs convolutional layers

11. CNN regularization
    - dropout
    - weight decay
    - augmentation
12. 1D CNN for biological sequences
    - sequence encoding
    - motif detection
    - local patterns
    - genomic classification
13. CNN vs RNN
    - local vs sequential processing
    - parallelism
    - receptive fields
14. CNN vs Transformer
    - locality
    - global context
    - computational trade-offs
15. CNN interpretability
    - learned filters
    - activation maps
    - feature visualization
16. CNN debugging
    - tensor shapes
    - channel ordering
    - padding mistakes

- overfitting

63. Natural Language Processing (NLP)

Preparedness target: STRONG

Your Search & Retrieval Engine makes NLP particularly relevant.

Subskills

1. Text preprocessing
    - normalization
    - punctuation
    - whitespace
    - case handling
2. Tokenization
    - word tokenization
    - subword tokenization
    - character tokenization
3. Stop words
    - stop-word removal
    - advantages
    - disadvantages
    - task dependence
4. Stemming
    - stemming
    - aggressive normalization
    - examples
5. Lemmatization

- linguistic normalization
- dictionary forms
- stemming vs lemmatization

6. Bag-of-words
    - vocabulary
    - term counts
    - sparse representation
7. TF-IDF
    - TF
    - IDF
    - document representation
    - similarity
8. N-grams
    - unigrams
    - bigrams
    - trigrams
    - phrase information
9. Word embeddings
    - dense vectors
    - semantic similarity
    - distributed representations
10. Word2Vec
    - CBOW
    - Skip-gram
    - negative sampling awareness
11. Contextual embeddings

- context-dependent representations
- transformer embeddings

12. Attention
    - query
    - key
    - value
    - attention weights
13. Transformer architecture
    - self-attention
    - positional information
    - encoder
    - decoder
14. Text classification
    - sentiment
    - topic classification
    - intent classification
15. Information retrieval
    - query processing
    - document retrieval
    - ranking
    - relevance
16. Semantic search
    - embeddings
    - similarity search
    - vector retrieval
17. Named Entity Recognition

- entity spans
- entity types
- sequence labelling

18. Sequence-to-sequence models
    - encoder-decoder
    - sequence generation
    - attention
19. NLP evaluation
    - precision
    - recall
    - F1
    - BLEU awareness
    - ROUGE awareness
20. NLP limitations
    - ambiguity
    - context
    - domain shift
    - vocabulary mismatch
21. NLP production pipeline
    - preprocessing
    - model inference
    - ranking
    - serving
    - monitoring

64. Embeddings & Vector Search

Preparedness target: STRONG

This overlaps with Sentence Transformers but is deliberately broader: you should understand the **system built around embeddings**, not just how to generate them.

Subskills

1. Embedding fundamentals
    - dense vectors
    - semantic representation
    - dimensionality
2. Embedding spaces
    - distance
    - similarity
    - neighbourhoods
    - semantic geometry
3. Similarity metrics
    - cosine similarity
    - Euclidean distance
    - dot product
4. Query embeddings
    - query encoding
    - document encoding
    - comparing representations
5. Vector databases
    - vector storage
    - vector indexing
    - nearest-neighbour retrieval
6. Approximate nearest-neighbour search

- ANN motivation
- exact vs approximate search
- recall vs latency

7. HNSW awareness
   - graph-based ANN
   - hierarchical layers
   - search trade-offs
8. Vector indexing
   - index construction
   - search parameters
   - memory trade-offs
9. Top-K retrieval
   - nearest neighbours
   - candidate generation
   - ranking
10. Hybrid retrieval
    - keyword search
    - BM25
    - vector search
    - score fusion
11. Embedding normalization
    - normalized vectors
    - cosine vs dot product
12. Embedding model selection
    - dimensionality
    - domain suitability

- latency
- quality

13. Embedding storage
    - memory requirements
    - serialization
    - indexing overhead
14. Retrieval evaluation
    - Recall@K
    - Precision@K
    - MRR
    - nDCG
15. Vector-search failure modes
    - semantic mismatch
    - irrelevant nearest neighbours
    - embedding drift
    - poor chunking
16. Retrieval pipeline design
    - ingestion
    - embedding
    - indexing
    - query encoding
    - retrieval
    - reranking

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65. Model Evaluation

Preparedness target: VERY STRONG

This deserves significant depth because interviewers frequently test whether candidates understand what their reported model metrics actually mean.

Subskills

1. Train/validation/test methodology
    - training set
    - validation set
    - test set
    - data leakage
2. Confusion matrix
    - true positive
    - true negative
    - false positive
    - false negative
3. Accuracy
    - definition
    - interpretation
    - limitations
4. Precision
    - definition
    - false positives
    - precision-oriented applications
5. Recall
    - definition
    - false negatives
    - recall-oriented applications

6. F1 score
    - harmonic mean
    - precision-recall trade-off
7. Macro vs micro averaging
    - macro metrics
    - micro metrics
    - class imbalance
8. ROC curve
    - TPR
    - FPR
    - thresholds
9. ROC-AUC
    - interpretation
    - ranking interpretation
    - limitations
10. Precision-recall curve
    - precision-recall trade-off
    - imbalanced datasets
11. PR-AUC
    - interpretation
    - comparison with ROC-AUC
12. Regression metrics
    - MAE
    - MSE
    - RMSE
    - R^2

13. Ranking metrics
    - Precision@K
    - Recall@K
    - MRR
    - nDCG
14. Calibration
    - predicted probabilities
    - calibration curves
    - reliability
15. Threshold selection
    - classification threshold
    - business trade-offs
    - precision/recall
16. Cross-validation
    - K-fold
    - stratified K-fold
    - time-aware validation
17. Statistical significance
    - variance
    - confidence intervals
    - repeated experiments
    - significance awareness
18. Error analysis
    - false positives
    - false negatives
    - difficult examples

- subgroup analysis

19. Baseline models
    - naive baseline
    - simple model
    - comparison against baseline
20. Model comparison
    - fair comparison
    - same dataset
    - same evaluation protocol
21. Evaluation leakage
    - test-set tuning
    - repeated test evaluation
    - hidden test contamination
22. Production evaluation
    - offline vs online metrics
    - monitoring
    - drift
    - feedback loops

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66. LIME

Preparedness target: INTERVIEW-READY / RESUME-DEFENSIBLE

Because you actually implemented LIME-style explainability, you should understand its mechanism rather than just its name.

Subskills

1. Model interpretability

- interpretability
- explainability
- black-box models

2. Local vs global explanations
    - local explanations
    - global model behavior
    - scope of explanations
3. LIME intuition
    - local approximation
    - perturbations
    - surrogate model
4. Input perturbation
    - generating samples
    - modifying features
    - neighborhood construction
5. Model predictions
    - black-box prediction function
    - evaluating perturbed inputs
6. Local surrogate model
    - simple interpretable model
    - fitting locally
    - feature weights
7. Sample weighting
    - locality
    - distance
    - weighting perturbations

8. Explanation generation
    - important features
    - positive contributions
    - negative contributions
9. LIME limitations
    - instability
    - locality
    - surrogate mismatch
    - perturbation assumptions
10. LIME vs SHAP
    - local explanations
    - theoretical differences
    - computational trade-offs
11. Explainability evaluation
    - fidelity
    - stability
    - interpretability

67. Data Cleaning & Preprocessing

Preparedness target: STRONG

Subskills

1. Data inspection
    - schema
    - data types
    - missing values

- duplicates

2. Missing-value handling
   - deletion
   - mean/median imputation
   - categorical imputation
   - model-based imputation
3. Outlier detection
   - IQR
   - z-score
   - domain-based detection
4. Duplicate detection
   - exact duplicates
   - near duplicates
   - identifier-based duplicates
5. Data-type conversion
   - numeric conversion
   - categorical conversion
   - dates
   - strings
6. Categorical encoding
   - one-hot encoding
   - ordinal encoding
   - target encoding awareness
7. Numerical scaling
   - standardization
   - min-max scaling

- robust scaling

8. Text preprocessing
    - normalization
    - tokenization
    - whitespace
    - encoding
9. Data leakage prevention
    - train/test separation
    - fitting preprocessors only on training data
10. Feature engineering
    - transformations
    - interactions
    - domain features
11. Feature selection
    - correlation
    - mutual information
    - model-based selection
12. Class imbalance
    - oversampling
    - undersampling
    - class weights
13. Data validation
    - schema validation
    - range checks
    - uniqueness checks
    - consistency checks

14. Preprocessing pipelines
    - reproducibility
    - train/inference consistency
    - sklearn pipelines

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68. Pandas

Preparedness target: STRONG

Pandas is very useful for your data-analysis/ML work and should be comfortable rather than merely familiar.

Subskills

1. Series and DataFrame
    - Series
    - DataFrame
    - index
    - columns
2. Data loading
    - CSV
    - JSON
    - Excel
    - SQL
3. Selection
    - loc
    - iloc
    - boolean filtering
    - column selection

4. Data transformation
    - assign
    - rename
    - map
    - apply
5. Missing data
    - isna
    - fillna
    - dropna
6. GroupBy
    - grouping
    - aggregation
    - transform
    - apply
7. Merge and join
    - merge
    - join
    - concatenation
    - key matching
8. Sorting
    - sort_values
    - sort_index
    - ranking
9. Reshaping
    - pivot
    - pivot_table

- melt
- stack/unstack

10. Time-series data
    - timestamps
    - date parsing
    - resampling
    - rolling windows
11. Vectorized operations
    - vectorization
    - avoiding unnecessary loops
    - performance
12. Data cleaning
    - duplicates
    - types
    - strings
    - missing values
13. Performance
    - memory usage
    - categorical types
    - chunked processing
14. Exporting data
    - CSV
    - JSON
    - Excel
    - database output

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69. NumPy

Preparedness target: STRONG

Subskills

1. ndarray
    - arrays
    - dimensions
    - shape
    - dtype
2. Array creation
    - zeros
    - ones
    - ranges
    - random arrays
3. Indexing and slicing
    - indexing
    - slicing
    - boolean masks
    - advanced indexing
4. Broadcasting
    - broadcasting rules
    - vectorized operations
5. Vectorization
    - array operations
    - avoiding Python loops
    - performance

6. Mathematical operations
    - aggregation
    - elementwise operations
    - linear algebra
7. Linear algebra
    - matrix multiplication
    - transpose
    - inverse awareness
    - decompositions awareness
8. Random numbers
    - random generation
    - seeds
    - distributions
9. Reshaping
    - reshape
    - flatten
    - squeeze
    - expand dimensions
10. Memory layout
    - contiguous arrays
    - views
    - copies
    - strides
11. NumPy performance
    - vectorization
    - memory locality

- avoiding unnecessary copies

12. NumPy vs Python lists
    - memory
    - speed
    - numerical operations

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70. Exploratory Data Analysis (EDA)

Preparedness target: STRONG

Subskills

1. Dataset overview
    - dimensions
    - columns
    - data types
    - missingness
2. Univariate analysis
    - distributions
    - central tendency
    - variance
    - outliers
3. Categorical analysis
    - frequency
    - cardinality
    - class distribution
4. Bivariate analysis
    - feature relationships

- correlation
- categorical vs numerical

5. Multivariate analysis
    - interactions
    - correlations
    - dimensionality
6. Correlation analysis
    - Pearson correlation
    - Spearman correlation
    - correlation limitations
7. Distribution analysis
    - skewness
    - heavy tails
    - transformations
8. Outlier analysis
    - statistical outliers
    - domain outliers
    - handling strategy
9. Missingness analysis
    - missing percentage
    - missingness patterns
    - MCAR/MAR/MNAR awareness
10. Class imbalance analysis
    - class frequencies
    - minority class
    - implications

11. Visualization
    - histograms
    - box plots
    - scatter plots
    - bar charts
    - heatmaps
12. Feature-target analysis
    - predictive relationships
    - leakage detection
    - feature usefulness
13. Data-quality analysis
    - inconsistent values
    - invalid ranges
    - duplicates
    - anomalies
14. EDA → modelling workflow
    - hypothesis formation
    - feature engineering
    - baseline modelling
    - iteration

71. pytest

Preparedness target: WORKING

Subskills

1. Test fundamentals

- test cases
- assertions
- expected behavior

2. pytest structure
    - test files
    - test functions
    - test discovery
3. Assertions
    - assert
    - expected exceptions
    - approximate values
4. Fixtures
    - fixture concept
    - setup
    - teardown
    - fixture scope
5. Parameterized tests
    - multiple inputs
    - edge cases
    - data-driven testing
6. Mocking
    - mocks
    - stubs
    - dependency isolation
7. API testing
    - endpoint tests

- request/response validation
- status codes

8. Integration testing
   - database integration
   - external dependencies
   - test environments
9. Test organization
   - unit vs integration
   - test naming
   - test isolation
10. Coverage
    - line coverage
    - branch coverage
    - coverage limitations

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72. JUnit 5

Preparedness target: WORKING

Subskills

1. JUnit fundamentals
   - test classes
   - test methods
   - assertions
2. Assertions
   - assertEquals
   - assertTrue

- assertThrows
- assertAll

3. Test lifecycle
   - BeforeEach
   - AfterEach
   - BeforeAll
   - AfterAll
4. Parameterized tests
   - parameter sources
   - repeated test cases
   - edge cases
5. Mockito awareness
   - mocks
   - stubs
   - verification
   - dependency isolation
6. Spring Boot testing
   - controller tests
   - service tests
   - repository tests
   - integration tests
7. Test organization
   - unit tests
   - integration tests
   - test naming
   - test isolation

8. Coverage and quality
    - code coverage
    - meaningful tests
    - regression tests

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73. Playwright

Preparedness target: WORKING

Subskills

1. Browser automation
    - browser
    - pages
    - contexts
    - automation
2. Locators
    - CSS selectors
    - text selectors
    - roles
    - stable locators
3. User interactions
    - click
    - type
    - select
    - keyboard
    - navigation
4. Assertions

- visibility
- text
- URL
- element state

5. End-to-end testing
    - user workflows
    - authentication
    - multi-page flows
6. Test isolation
    - browser contexts
    - test data
    - cleanup
7. Waiting strategies
    - auto-waiting
    - explicit waits
    - avoiding flaky sleeps
8. Screenshots and debugging
    - screenshots
    - traces
    - video
    - failure debugging
9. Playwright architecture
    - browser
    - context
    - page
    - test runner

10. E2E test design
    - critical paths
    - regression coverage
    - stable tests
    - flaky-test prevention

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74. CI/CD

Preparedness target: STRONG

You should understand the engineering process, not just know what CI/CD stands for.

Subskills

1. Continuous Integration
    - frequent integration
    - automated builds
    - automated tests
2. Continuous Delivery
    - deployable artifacts
    - release readiness
    - approval gates
3. Continuous Deployment
    - automatic production deployment
    - deployment triggers
4. Pipeline stages
    - checkout
    - install

- build
- test
- package
- deploy

5. Build automation
    - reproducible builds
    - dependency installation
    - artifacts
6. Automated testing
    - unit tests
    - integration tests
    - E2E tests
7. Quality gates
    - tests
    - linting
    - coverage
    - security checks
8. Secrets management
    - environment secrets
    - credentials
    - avoiding hardcoded secrets
9. Deployment strategies
    - rolling deployment
    - blue-green awareness
    - canary awareness
10. Rollbacks

- failed deployment
- previous version
- rollback strategy

11. Environment management
    - development
    - staging
    - production
    - configuration differences
12. Build artifacts
    - artifact creation
    - artifact storage
    - deployment artifacts
13. CI/CD failure debugging
    - logs
    - failed tests
    - dependency failures
    - environment mismatch
14. CI/CD security
    - secret protection
    - dependency scanning
    - permissions
    - supply-chain awareness
15. Pipeline optimization
    - caching
    - parallel jobs
    - incremental builds

- reducing build time

75. GitHub Actions

Preparedness target: WORKING

Subskills

1. Workflow fundamentals
    - workflow files
    - YAML
    - triggers
    - jobs
    - steps
2. Events
    - push
    - pull request
    - manual dispatch
    - scheduled workflows
3. Jobs
    - runners
    - dependencies
    - parallel jobs
    - conditional execution
4. Steps
    - actions
    - shell commands
    - environment variables

5. Secrets
    - repository secrets
    - environment secrets
    - secure credential usage
6. CI pipelines
    - install dependencies
    - build
    - test
    - lint
7. Artifacts
    - uploading artifacts
    - downloading artifacts
    - test reports
8. Caching
    - dependency caching
    - cache keys
    - build acceleration
9. Matrix builds
    - multiple versions
    - multiple environments
    - compatibility testing
10. Deployment workflows
    - build
    - deploy
    - environment protection
11. GitHub Actions debugging

- workflow logs
- failed steps
- environment differences
- permissions

12. GitHub Actions security
    - least-privilege permissions
    - secrets
    - untrusted pull requests
    - dependency actions

76. Vercel

Preparedness target: BASIC / WORKING

Subskills

1. Vercel fundamentals
    - deployment platform
    - projects
    - deployments
    - environments
2. Frontend deployment
    - production builds
    - static assets
    - framework integration
3. Preview deployments
    - preview environments
    - pull-request deployments

- deployment URLs

4. Environment variables
    - development variables
    - preview variables
    - production variables
    - secrets
5. Build configuration
    - build commands
    - output directories
    - framework detection
6. Domains
    - custom domains
    - DNS awareness
    - HTTPS
7. Serverless/API functions awareness
    - request handlers
    - execution model
    - limitations
8. Deployment debugging
    - build logs
    - runtime logs
    - environment mismatches
    - failed deployments
9. Vercel architecture trade-offs
    - serverless vs traditional servers
    - cold starts

- scaling
- vendor lock-in

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77. Render

Preparedness target: BASIC / WORKING

Subskills

1. Render fundamentals
    - services
    - deployments
    - environments
2. Web-service deployment
    - build command
    - start command
    - ports
    - environment variables
3. Backend deployment
    - Node/Express
    - Flask/FastAPI
    - Java services
4. Database deployment awareness
    - managed databases
    - connection strings
    - credentials
5. Environment configuration
    - environment variables

- secrets
- production configuration

6. Deployment lifecycle
    - build
    - deploy
    - restart
    - logs
7. Debugging deployments
    - build failures
    - runtime failures
    - port issues
    - environment problems
8. Render vs traditional hosting
    - managed deployment
    - infrastructure abstraction
    - scaling limitations

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78. Supabase

Preparedness target: STRONG

Supabase is now particularly important because **THE SYSTEM itself uses Supabase**.

Subskills

1. Supabase architecture
    - hosted PostgreSQL
    - database

- authentication
- storage
- APIs

2. PostgreSQL integration
    - tables
    - schemas
    - SQL
    - relationships
3. Supabase Auth
    - users
    - sessions
    - authentication
    - providers
    - tokens
4. Row Level Security
    - RLS
    - policies
    - user-based access
    - authorization
5. Supabase API
    - generated APIs
    - CRUD operations
    - filtering
    - relationships
6. Database relationships
    - foreign keys

- one-to-many
- many-to-many
- referential integrity

7. Supabase migrations
    - schema migrations
    - version control
    - migration ordering
    - rollback awareness
8. Database functions
    - stored functions
    - RPC
    - server-side logic
9. Supabase storage
    - buckets
    - file upload
    - access policies
    - public/private files
10. Authentication + frontend integration
    - login
    - logout
    - session persistence
    - protected routes
11. Supabase security
    - RLS
    - service-role keys
    - anon keys

- secret management

12. Supabase performance
    - indexes
    - query optimization
    - connection considerations
    - pagination
13. Supabase schema design
    - normalized data
    - ownership
    - global vs user-specific entities
    - soft deletion
14. Supabase architecture for THE SYSTEM
    - global skill catalogue
    - custom skills
    - subskills
    - user mastery
    - quiz questions
    - quiz attempts
    - notes
    - company associations
15. Supabase failure/debugging
    - authentication failures
    - RLS failures
    - API errors
    - migration errors

79. Cloud Computing Fundamentals

Preparedness target: STRONG

Subskills

1. Cloud-computing concepts
   - on-demand resources
   - elasticity
   - scalability
   - managed services
2. IaaS
   - virtual machines
   - networking
   - storage
   - infrastructure management
3. PaaS
   - managed application platforms
   - deployment abstraction
   - runtime environments
4. SaaS
   - software as a service
   - managed applications
   - subscription model
5. Virtual machines
   - hypervisors
   - virtual CPUs
   - virtual memory

- images

6. Containers vs VMs
    - isolation
    - startup time
    - resource overhead
    - use cases
7. Object storage
    - buckets
    - objects
    - metadata
    - durability
8. Block storage
    - disks
    - volumes
    - filesystem interaction
9. Cloud networking
    - private networks
    - subnets
    - public/private resources
    - security groups
10. Load balancing
    - traffic distribution
    - health checks
    - failover
11. Autoscaling
    - horizontal scaling

- vertical scaling
- scaling triggers
- capacity planning

12. Availability zones
    - fault domains
    - redundancy
    - high availability
13. Managed databases
    - backups
    - replication
    - scaling
    - operational trade-offs
14. Cloud security
    - IAM
    - least privilege
    - secrets
    - network isolation
15. Cloud cost awareness
    - compute cost
    - storage cost
    - bandwidth
    - managed-service trade-offs
16. Cloud architecture trade-offs
    - managed vs self-hosted
    - cost vs reliability
    - simplicity vs control

80. Kubernetes

Preparedness target: INTERVIEW-READY

You don't need Kubernetes administrator-level expertise in 100 days. But for SDE/backend roles, you should understand the architecture.

Subskills

1. Kubernetes fundamentals
    - cluster
    - control plane
    - worker nodes
    - workloads
2. Pods
    - pod
    - containers
    - shared networking
    - pod lifecycle
3. Deployments
    - desired state
    - replicas
    - rolling updates
    - rollback
4. Services
    - service abstraction
    - stable networking
    - service discovery

- ClusterIP

5. Ingress
   - HTTP routing
   - host/path routing
   - ingress controller awareness
6. ConfigMaps
   - configuration
   - environment variables
   - non-secret settings
7. Secrets
   - credentials
   - secret injection
   - security considerations
8. Scaling
   - replicas
   - horizontal scaling
   - autoscaling awareness
9. Health checks
   - liveness probes
   - readiness probes
   - startup probes
10. Kubernetes networking
    - pod networking
    - service networking
    - DNS
    - network policies awareness

11. Storage
    - volumes
    - persistent volumes
    - persistent volume claims
12. Scheduling
    - nodes
    - resource requests
    - resource limits
    - scheduling decisions
13. Kubernetes debugging
    - logs
    - describe
    - events
    - failed pods
    - networking issues
14. Kubernetes architecture trade-offs
    - orchestration
    - complexity
    - scalability
    - operational overhead

---

81. System Design

Preparedness target: VERY STRONG

This is one of the most important placement skills in the entire catalogue.

Subskills

1. Requirements gathering
    - functional requirements
    - non-functional requirements
    - assumptions
    - constraints
2. Capacity estimation
    - users
    - requests per second
    - storage
    - bandwidth
    - peak traffic
3. API design
    - endpoints
    - request/response
    - idempotency
    - versioning
4. Data modelling
    - entities
    - relationships
    - schema design
    - normalization
5. Architecture decomposition
    - services
    - responsibilities
    - boundaries
    - dependencies

6. Monolith vs microservices
    - deployment
    - scaling
    - complexity
    - organizational trade-offs
7. Load balancing
    - horizontal scaling
    - routing
    - health checks
8. Caching
    - cache-aside
    - invalidation
    - TTL
    - cache failures
9. Database scaling
    - read replicas
    - sharding
    - partitioning
    - indexing
10. Asynchronous architecture
    - queues
    - Kafka
    - event-driven systems
    - background processing
11. Reliability
    - redundancy

- failover
- retries
- graceful degradation

12. Rate limiting
    - fixed window
    - sliding window
    - token bucket
    - distributed rate limiting
13. Consistency
    - strong consistency
    - eventual consistency
    - read-after-write
14. Availability
    - uptime
    - redundancy
    - failover
    - disaster recovery awareness
15. Distributed-system failure
    - partial failures
    - network partitions
    - cascading failures
16. Observability
    - logs
    - metrics
    - traces
    - alerts

17. Security
    - authentication
    - authorization
    - encryption
    - secrets
18. Performance
    - latency
    - throughput
    - bottlenecks
    - tail latency
19. Scalability
    - vertical
    - horizontal
    - stateless architecture
    - partitioning
20. System-design trade-offs
    - cost
    - complexity
    - reliability
    - performance
    - consistency
21. Common system-design problems
    - URL shortener
    - social feed
    - chat
    - notification service

- file storage
- rate limiter
- distributed cache
- search engine

---

82. Operating Systems

Preparedness target: VERY STRONG

This is core CS. It should receive substantial preparation even if you don't particularly enjoy systems.

Subskills

1. OS fundamentals
    - kernel
    - user space
    - system calls
    - hardware interaction
2. Processes
    - process
    - PCB
    - process states
    - process creation
    - process termination
3. Threads
    - user threads
    - kernel threads
    - process/thread relationship

4. Context switching
    - CPU state
    - context switch
    - overhead
5. CPU scheduling
    - FCFS
    - SJF
    - SRTF
    - Round Robin
    - priority scheduling
6. Scheduling metrics
    - waiting time
    - turnaround time
    - response time
    - throughput
7. Inter-process communication
    - pipes
    - shared memory
    - message queues
    - sockets
8. Synchronization
    - critical sections
    - mutex
    - semaphore
    - monitor
    - condition variable

9. Deadlocks
    - four necessary conditions
    - prevention
    - avoidance
    - detection
    - recovery
10. Memory management
    - physical memory
    - virtual memory
    - address spaces
    - allocation
11. Paging
    - pages
    - frames
    - page tables
    - address translation
12. TLB
    - translation lookaside buffer
    - address-translation caching
    - TLB misses
13. Page faults
    - demand paging
    - page-fault handling
    - disk access
14. Page-replacement algorithms
    - FIFO

    - LRU
    - Optimal
    - Clock
15. Virtual memory
    - isolation
    - paging
    - swapping
    - working sets
16. Memory fragmentation
    - internal fragmentation
    - external fragmentation
17. File systems
    - files
    - directories
    - metadata
    - file descriptors
18. File-system allocation
    - contiguous allocation
    - linked allocation
    - indexed allocation
19. Inodes
    - inode
    - metadata
    - block pointers
    - directories
20. Disk scheduling

- FCFS
- SSTF
- SCAN
- C-SCAN

21. System calls
    - process calls
    - file calls
    - memory calls
    - networking calls
22. User/kernel mode
    - privilege levels
    - mode switching
    - protection
23. Interrupts
    - hardware interrupts
    - software interrupts
    - interrupt handling
24. OS security
    - isolation
    - permissions
    - privilege
    - access control

---

83. Computer Networks

Preparedness target: VERY STRONG

This is another core placement subject.

Subskills

1. Network models
    - OSI
    - TCP/IP
    - layer responsibilities
2. Ethernet
    - frames
    - MAC addresses
    - switching
3. ARP
    - IP-to-MAC resolution
    - ARP cache
4. IP addressing
    - IPv4
    - IPv6 awareness
    - addresses
    - subnets
5. CIDR
    - subnet masks
    - prefix notation
    - address ranges
6. Routing
    - routers
    - routing tables

- next hop

7. NAT
    - private addresses
    - public addresses
    - port translation
8. TCP
    - connection establishment
    - reliability
    - flow control
    - congestion control
9. UDP
    - datagrams
    - connectionless communication
    - trade-offs
10. TCP vs UDP
    - reliability
    - latency
    - ordering
    - use cases
11. DNS
    - domain resolution
    - recursive resolver
    - authoritative server
    - caching
12. HTTP
    - request/response

- methods
- headers
- status codes

13. HTTP/1.1
    - persistent connections
    - pipelining awareness
14. HTTP/2
    - multiplexing
    - binary framing
    - streams
15. HTTP/3
    - QUIC awareness
    - UDP foundation
    - connection establishment
16. TLS/HTTPS
    - encryption
    - certificates
    - handshake
    - authentication
17. Sockets
    - IP
    - port
    - socket
    - client/server communication
18. Network congestion
    - congestion

- packet loss
- throughput
- latency

19. Flow control
    - receiver capacity
    - TCP windows
20. Network reliability
    - retransmission
    - acknowledgements
    - timeout
21. Network debugging
    - ping
    - traceroute
    - DNS lookup
    - port inspection
    - packet capture awareness
22. Proxies
    - forward proxy
    - reverse proxy
    - gateway
23. Load balancers
    - L4
    - L7
    - routing
24. Network security
    - firewalls

- TLS
- authentication
- attacks awareness

84. DBMS

Preparedness target: VERY STRONG

This should be one of your highest-priority interview subjects.

Subskills

1. Relational databases
    - relations
    - tuples
    - attributes
    - schemas
2. Keys
    - primary key
    - candidate key
    - superkey
    - foreign key
3. Functional dependencies
    - dependencies
    - closure
    - normalization reasoning
4. Normalization
    - 1NF
    - 2NF

- 3NF
- BCNF

5. Denormalization
    - redundancy
    - read optimization
    - write trade-offs
6. SQL
    - querying
    - joins
    - aggregation
    - subqueries
7. Transactions
    - transaction boundaries
    - commit
    - rollback
8. ACID
    - atomicity
    - consistency
    - isolation
    - durability
9. Isolation levels
    - read uncommitted
    - read committed
    - repeatable read
    - serializable
10. Concurrency anomalies

- dirty read
- non-repeatable read
- phantom read
- lost update

11. Locks
    - shared locks
    - exclusive locks
    - lock compatibility
12. Deadlocks
    - transaction deadlock
    - detection
    - prevention
    - recovery
13. MVCC
    - multi-version concurrency
    - snapshots
    - visibility
14. Indexes
    - B-tree
    - hash
    - composite indexes
    - covering indexes
15. Query processing
    - parsing
    - optimization
    - execution

16. Query plans
    - scans
    - joins
    - cost estimation
17. Storage
    - pages
    - records
    - buffer pool
18. Buffer management
    - memory caching
    - dirty pages
    - eviction
19. Write-ahead logging
    - WAL
    - durability
    - recovery
20. Crash recovery
    - logs
    - checkpoints
    - redo
    - undo awareness
21. Replication
    - primary-replica
    - synchronous
    - asynchronous
22. Sharding

- horizontal partitioning
- shard keys
- hotspots

23. NoSQL awareness
    - key-value
    - document
    - column-family
    - graph databases
24. OLTP vs OLAP
    - transactions
    - analytics
    - workload characteristics
25. Database design trade-offs
    - consistency
    - performance
    - normalization
    - scalability

---

85. Object-Oriented Programming

Preparedness target: VERY STRONG

Subskills

1. Classes and objects
   - class
   - object
   - state

- behavior

2. Encapsulation
   - information hiding
   - access control
   - interfaces
3. Abstraction
   - abstract interfaces
   - implementation hiding
4. Inheritance
   - base class
   - derived class
   - reuse
   - inheritance problems
5. Polymorphism
   - compile-time polymorphism
   - runtime polymorphism
6. Method overloading
   - signatures
   - compile-time resolution
7. Method overriding
   - runtime dispatch
   - virtual methods
8. Composition
   - has-a relationship
   - object composition
9. Composition vs inheritance

- coupling
- reuse
- maintainability

10. Interfaces
    - contracts
    - implementations
    - dependency inversion
11. Abstract classes
    - abstract methods
    - partial implementation
12. Constructors/destructors
    - object lifecycle
    - initialization
    - cleanup
13. SOLID principles
    - Single Responsibility
    - Open/Closed
    - Liskov Substitution
    - Interface Segregation
    - Dependency Inversion
14. Design principles
    - DRY
    - KISS
    - separation of concerns
    - low coupling
    - high cohesion

15. OOP design trade-offs
    - abstraction vs complexity
    - inheritance vs composition
    - flexibility vs simplicity

---

86. Design Patterns

Preparedness target: STRONG

Don't memorize 30 patterns. Understand the common ones and when they are appropriate.

Subskills

1. Singleton
    - single instance
    - global state problems
    - thread safety
2. Factory
    - object creation
    - decoupling construction
3. Abstract Factory
    - families of objects
    - abstract creation
4. Builder
    - complex object construction
    - optional parameters
5. Adapter
    - interface compatibility

- wrapping existing components

6. Decorator
    - dynamic behavior
    - composition
7. Facade
    - simplified interface
    - subsystem abstraction
8. Proxy
    - controlled access
    - lazy loading
    - remote proxy
9. Observer
    - event notification
    - publisher/subscriber
10. Strategy
    - interchangeable algorithms
    - runtime selection
11. Command
    - encapsulated operations
    - queues
    - undo awareness
12. Template Method
    - algorithm skeleton
    - overridable steps
13. Repository pattern
    - data-access abstraction

- persistence separation

14. Dependency Injection
    - dependency inversion
    - testability
15. Pattern selection
    - recognizing problems
    - avoiding unnecessary patterns
    - trade-offs

87. Data Structures

Preparedness target: VERY STRONG

This is foundational for your DSA preparation.

Subskills

1. Arrays
    - indexing
    - insertion/deletion
    - contiguous memory
    - complexity
2. Strings
    - representation
    - manipulation
    - pattern matching
3. Linked lists
    - singly linked
    - doubly linked

- insertion/deletion
- pointer manipulation

4. Stacks
   - LIFO
   - applications
   - implementation
5. Queues
   - FIFO
   - circular queue
   - applications
6. Deques
   - double-ended operations
   - monotonic deque
7. Hash tables
   - hashing
   - collisions
   - chaining
   - open addressing
8. Trees
   - binary trees
   - traversals
   - height
   - recursion
9. BST
   - search
   - insertion

- deletion
- ordering

10. Balanced trees
    - AVL awareness
    - Red-Black awareness
11. Heaps
    - min heap
    - max heap
    - heapify
    - priority queues
12. Tries
    - prefix trees
    - insertion
    - search
    - prefix queries
13. Graph representations
    - adjacency matrix
    - adjacency list
    - edge list
14. Disjoint Set Union
    - union
    - find
    - path compression
    - union by rank/size
15. Segment trees
    - range queries

- range updates
- lazy propagation

16. Fenwick trees
    - prefix sums
    - point updates
    - binary indexed tree
17. Sparse tables
    - static range queries
    - preprocessing
    - idempotent operations
18. Complexity analysis
    - time complexity
    - space complexity
    - amortized complexity

---

88. Algorithms

Preparedness target: VERY STRONG

This should be one of the highest-weighted skills in your entire tracker.

Subskills

1. Complexity analysis
    - Big-O
    - Big-Theta
    - Big-Omega
    - amortized complexity
2. Sorting

- bubble sort
- selection sort
- insertion sort
- merge sort
- quicksort
- heap sort

3. Searching
    - linear search
    - binary search
    - search-space reduction
4. Two pointers
    - opposing pointers
    - same-direction pointers
    - sorted arrays
5. Sliding window
    - fixed window
    - variable window
    - frequency tracking
6. Prefix sums
    - 1D
    - 2D
    - range queries
7. Difference arrays
    - range updates
    - reconstruction
8. Hashing techniques

- frequency maps
- complement lookup
- state compression

9. Greedy algorithms
   - local optimality
   - exchange arguments
   - interval scheduling
10. Divide and conquer
    - recursive decomposition
    - merge
    - recurrence analysis
11. Backtracking
    - state space
    - pruning
    - permutations
    - combinations
12. Dynamic programming
    - state definition
    - recurrence
    - memoization
    - tabulation
    - space optimization
13. Knapsack patterns
    - 0/1 knapsack
    - unbounded knapsack
    - subset sum

14. LIS/LCS patterns
    - LIS
    - LCS
    - edit distance
15. Tree algorithms
    - DFS
    - BFS
    - diameter
    - subtree problems
    - tree DP
16. Graph traversal
    - BFS
    - DFS
    - connected components
17. Shortest paths
    - BFS
    - Dijkstra
    - Bellman-Ford
    - Floyd-Warshall
18. MST
    - Kruskal
    - Prim
    - DSU
19. Topological sorting
    - Kahn's algorithm
    - DFS ordering

- DAGs

20. Strongly connected components
    - Kosaraju
    - Tarjan awareness
21. Bipartite graphs
    - coloring
    - BFS/DFS
    - odd cycles
22. Advanced graph algorithms
    - bridges
    - articulation points
    - Eulerian paths
    - network-flow awareness
23. String algorithms
    - KMP
    - Z algorithm
    - rolling hash
    - trie
24. Monotonic structures
    - monotonic stack
    - monotonic queue
    - next greater element
25. Bit manipulation
    - bit masks
    - XOR
    - shifts

- subset enumeration

26. Number theory
    - GCD
    - LCM
    - sieve
    - modular arithmetic
    - fast exponentiation
27. Binary search on answer
    - feasibility function
    - monotonic predicate
    - optimization problems
28. Meet-in-the-middle
    - state splitting
    - subset enumeration
    - combining halves
29. Randomized algorithms
    - randomization
    - expected complexity
    - randomized selection
30. Competitive-programming problem solving
    - recognizing patterns
    - constraint analysis
    - implementation strategy
    - edge cases

---

89. Linear Programming

Preparedness target: INTERVIEW-READY / STRONG

This is especially relevant to your research background.

Subskills

1. Optimization fundamentals
   - objective
   - variables
   - constraints
   - feasible solutions
2. Linear programs
   - linear objective
   - linear constraints
   - decision variables
3. Standard form
   - equality constraints
   - non-negativity
   - canonical representations
4. Feasible region
   - feasible set
   - vertices
   - boundedness
   - infeasibility
5. Optimal solutions
   - objective maximization/minimization
   - optimal vertex

6. Slack variables
    - inequalities
    - conversion to equalities
7. Simplex method
    - basic feasible solutions
    - pivots
    - entering/leaving variables
    - simplex intuition
8. Degeneracy
    - degenerate solutions
    - cycling awareness
9. Unboundedness
    - unbounded objective
    - detecting unbounded problems
10. Infeasibility
    - conflicting constraints
    - infeasible region
11. Duality awareness
    - primal
    - dual
    - relationship
12. LP applications
    - resource allocation
    - scheduling
    - transportation
    - assignment

90. Linear Programming Duality

Preparedness target: STRONG

Subskills

1. Primal and dual
    - primal formulation
    - dual formulation
    - correspondence
2. Weak duality
    - primal objective
    - dual objective
    - bound relationship
3. Strong duality
    - optimal primal value
    - optimal dual value
4. Dual variables
    - shadow prices
    - marginal value
5. Complementary slackness
    - primal constraints
    - dual variables
    - optimality conditions
6. Economic interpretation
    - resource value
    - opportunity cost
    - marginal value

7. Duality in optimization
    - certificates
    - bounds
    - alternative formulations
8. Duality applications
    - resource allocation
    - scheduling
    - security/resource games

---

91. Integer Programming

Preparedness target: WORKING / INTERVIEW-READY

Subskills

1. Integer programming
    - integer variables
    - discrete decisions
2. Binary variables
    - 0/1 decisions
    - selection
    - assignment
3. Mixed-integer programming
    - continuous variables
    - integer variables
    - combined models
4. LP relaxation
    - relaxing integrality

- bounds
- fractional solutions

5. Branch and bound
    - branching
    - bounds
    - pruning
6. Cutting planes
    - valid inequalities
    - tightening relaxations
7. ILP applications
    - scheduling
    - allocation
    - routing
    - selection
8. Computational complexity
    - NP-hardness awareness
    - scalability
    - solver trade-offs

---

92. Game Theory

Preparedness target: STRONG / SPECIALIZED

This is particularly important because of your academic work.

Subskills

1. Game-theory fundamentals
    - players

- strategies
- payoffs
- preferences

2. Normal-form games
   - payoff matrices
   - pure strategies
   - mixed strategies
3. Best response
   - best-response correspondence
   - strategic reasoning
4. Dominant strategies
   - strictly dominant
   - weakly dominant
5. Nash equilibrium
   - definition
   - mutual best response
   - pure-strategy equilibrium
6. Mixed-strategy Nash equilibrium
   - probabilities
   - indifference principle
   - expected utility
7. Zero-sum games
   - opposing payoffs
   - minimax reasoning
8. Minimax theorem
   - maximin

- minimax
- zero-sum equilibrium

9. Extensive-form games
    - game trees
    - actions
    - information sets
10. Perfect information
    - observable actions
    - sequential games
11. Imperfect information
    - information sets
    - hidden actions
    - uncertainty
12. Sequential games
    - backward induction
    - subgame reasoning
13. Bayesian games
    - types
    - beliefs
    - incomplete information
14. Mechanism design awareness
    - incentives
    - truthful mechanisms
    - social objectives
15. Evolutionary game theory awareness
    - populations

- evolutionary stability
- strategy dynamics

16. Game-theory applications
    - security
    - resource allocation
    - auctions
    - markets

---

93. Stackelberg Games

Preparedness target: VERY STRONG / SPECIALIZED

This is directly relevant to your seminar and research.

Subskills

1. Stackelberg game fundamentals
    - leader
    - follower
    - sequential decisions
2. Leader commitment
    - commitment
    - strategy selection
    - follower response
3. Follower best response
    - optimization
    - response correspondence
4. Stackelberg equilibrium
    - leader optimization

- follower response
- equilibrium

5. Mixed strategies
    - probability distributions
    - randomized commitment
6. Resource allocation
    - limited resources
    - allocation decisions
    - utility
7. Security-game formulation
    - defender
    - attacker
    - targets
    - coverage
8. Security games
    - security resources
    - attack probabilities
    - protection
9. Stackelberg Security Games
    - defender commitment
    - attacker response
    - optimal coverage
10. Coverage strategies
    - marginal coverage
    - allocation
    - feasibility

11. Multiple targets
    - target values
    - attack probabilities
    - resource constraints
12. Uncertainty
    - attacker types
    - incomplete information
    - uncertainty over preferences
13. Optimization formulation
    - objective
    - constraints
    - equilibrium constraints
14. Computational challenges
    - large action spaces
    - scalability
    - optimization complexity
15. Security-game applications
    - patrol
    - infrastructure protection
    - cybersecurity
    - defense planning

---

94. Stackelberg Security Games

Preparedness target: VERY STRONG / SPECIALIZED

This gets its own skill because it is more specific than general Stackelberg theory.

Subskills

1. Defender-attacker model
   - defender
   - attacker
   - targets
   - payoffs
2. Coverage probability
   - marginal coverage
   - protection probability
   - resource allocation
3. Attacker utility
   - attack reward
   - attack cost
   - expected utility
4. Defender utility
   - protection value
   - expected loss
   - resource constraints
5. Attacker best response
   - target selection
   - expected utility comparison
6. Mixed defender strategy
   - randomized allocation
   - marginal probabilities
7. Multiple attacker types
   - attacker types

- type probabilities
- Bayesian response

8. Security-game LP formulations
    - optimization
    - coverage constraints
    - utility constraints
9. SSE
    - Strong Stackelberg Equilibrium
    - tie-breaking
    - leader-favorable response
10. Resource constraints
    - limited defenders
    - target coverage
    - feasibility
11. Real-world security applications
    - patrol scheduling
    - airport security
    - infrastructure
    - cyber defense
12. SAM/SEAD application
    - SAM resources
    - routes
    - attacker aircraft
    - defense allocation
    - expected payoff
13. Model limitations

- assumptions
- payoff estimation
- behavioral uncertainty
- computational scalability

---

95. CFR — Counterfactual Regret Minimization

Preparedness target: VERY STRONG / SPECIALIZED

This is directly connected to your seminar.

Subskills

1. Regret minimization
   - regret
   - cumulative regret
   - external regret
2. Counterfactual reasoning
   - counterfactual value
   - information sets
   - reach probabilities
3. Extensive-form games
   - game tree
   - information sets
   - imperfect information
4. CFR intuition
   - regret at information sets
   - iterative strategy improvement

5. Counterfactual value
    - expected utility
    - reach probability
    - opponent reach
6. Regret update
    - instantaneous regret
    - cumulative regret
    - positive regret
7. Strategy construction
    - regret matching
    - action probabilities
    - normalization
8. Average strategy
    - strategy averaging
    - convergence interpretation
9. CFR iteration
    - traversal
    - utility calculation
    - regret update
    - strategy update
10. CFR convergence
    - regret bound
    - approximate Nash equilibrium
11. CFR variants
    - CFR+
    - Monte Carlo CFR awareness

12. CFR complexity
    - game-tree size
    - information-set count
    - computational cost
13. CFR applications
    - poker
    - imperfect-information games
    - security games awareness

---

96. Regret Minimization & Online Learning

Preparedness target: VERY STRONG / SPECIALIZED

Subskills

1. Online learning
    - rounds
    - actions
    - losses
    - feedback
2. Regret
    - cumulative regret
    - average regret
    - comparator strategy
3. External regret
    - best fixed action
    - regret definition
4. Internal/swap regret awareness

- action-dependent regret
- stronger guarantees

5. Full-information feedback
   - observing all losses
   - online optimization
6. Partial-information feedback
   - bandit feedback
   - observing selected action
7. Exploration vs exploitation
   - exploration
   - exploitation
   - trade-off
8. Multiplicative weights
   - weight updates
   - exponential updates
   - regret guarantees
9. Hedge
   - exponential weighting
   - learning rate
10. Online gradient descent awareness
    - gradients
    - convex losses
    - iterative updates
11. Regret bounds
    - sublinear regret
    - average regret approaching zero

12. Adversarial environments
    - adaptive opponents
    - worst-case reasoning
13. Bandit learning
    - arm selection
    - reward observation
    - exploration
14. Online learning applications
    - recommendation
    - resource allocation
    - security

97. Stackelberg Security Games with Learning

Preparedness target: VERY STRONG / SPECIALIZED

This captures the specific research direction you've been studying rather than merely generic game theory.

Subskills

1. Repeated security games
    - repeated interaction
    - changing attacker behavior
2. Unknown attacker response
    - unknown preferences
    - learning response models
3. Online Stackelberg optimization
    - repeated decisions

- feedback
- regret

4. Commitment without regret
   - commitment strategy
   - online learning
   - regret minimization
5. Frequency vectors
   - empirical action frequencies
   - strategy representation
6. Barycentric spanners
   - basis vectors
   - convex representation
   - approximation
7. Exploration/exploitation blocks
   - exploration phases
   - exploitation phases
   - block structure
8. Partial-information learning
   - bandit feedback
   - observed losses
   - uncertainty
9. Full-information learning
   - complete feedback
   - comparison against all actions
10. Regret guarantees
    - sublinear regret

- convergence interpretation
- dependence on horizon

11. T^(2/3)-type regret
    - exploration/exploitation balance
    - partial-information trade-off
    - asymptotic guarantee
12. Learning in security games
    - unknown attacker
    - adaptive defender
    - empirical learning

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98. Patrol Theory / Security Resource Allocation

Preparedness target: STRONG / RESEARCH-SPECIFIC

This is directly useful for your current research work.

Subskills

1. Patrol planning
    - patrol routes
    - resources
    - coverage
2. Resource allocation
    - limited resources
    - target priorities
    - allocation constraints
3. Target prioritization

- high/medium/low value
- threat levels
- protection priorities

4. Coverage probabilities
    - patrol frequency
    - probability of interception
    - resource distribution
5. Attacker route selection
    - routes
    - target selection
    - expected payoff
6. Defender utility
    - protected targets
    - losses
    - expected utility
7. Attacker utility
    - successful attack
    - interception
    - route value
8. Optimization formulation
    - decision variables
    - constraints
    - objective
9. Game-theoretic patrol planning
    - defender strategy
    - attacker response

- equilibrium

10. Practical model conversion
    - mathematical model
    - implementable inputs
    - simulation
    - measurable outputs
11. Model assumptions
    - rationality
    - observability
    - payoff assumptions
    - resource assumptions
12. Simulation
    - scenarios
    - repeated trials
    - empirical outcomes
    - sensitivity analysis

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99. Mathematical Modelling & Optimization

Preparedness target: STRONG

You don't need to become a mathematician, but you should be able to translate a real problem into an implementable optimization model.

Subskills

1. Problem abstraction
    - real-world problem
    - assumptions

- decision variables

2. Decision variables
    - continuous
    - integer
    - binary
3. Objective functions
    - maximize
    - minimize
    - multi-objective awareness
4. Constraints
    - resource
    - capacity
    - logical
    - feasibility
5. Linear models
    - linear objective
    - linear constraints
6. Integer models
    - discrete decisions
    - binary variables
7. Nonlinear models
    - nonlinear objective
    - nonlinear constraints
    - awareness
8. Feasibility
    - feasible solution

- infeasible model
- constraint conflicts

9. Sensitivity analysis
    - parameter changes
    - solution stability
    - robustness
10. Trade-off analysis
    - competing objectives
    - Pareto frontier awareness
11. Model validation
    - assumptions
    - expected behavior
    - simulation
    - sanity checks
12. Mathematical model → software
    - input representation
    - solver interface
    - output interpretation
13. Optimization engineering
    - solver selection
    - runtime
    - numerical issues
    - scalability

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100. Research & Experimental Methodology

Preparedness target: STRONG

This matters because you are an M.Tech student and have to defend research/project work in interviews.

Subskills

1. Problem formulation
    - research question
    - hypotheses
    - assumptions
2. Literature review
    - finding related work
    - identifying gaps
    - comparing approaches
3. Baselines
    - simple baseline
    - existing method
    - fair comparison
4. Experimental design
    - variables
    - controlled experiments
    - datasets
    - scenarios
5. Evaluation metrics
    - appropriate metrics
    - quantitative comparison
    - qualitative analysis
6. Ablation studies

- component removal
- contribution analysis

7. Sensitivity analysis
    - parameter variation
    - robustness
8. Reproducibility
    - seeds
    - configuration
    - environment
    - code/data
9. Statistical reasoning
    - mean
    - variance
    - confidence intervals
    - significance awareness
10. Benchmarking
    - workload
    - baseline
    - measurement methodology
    - repeated trials
11. Error analysis
    - failure cases
    - systematic errors
    - qualitative investigation
12. Scientific communication
    - methodology

- results
- limitations
- conclusions

13. Research limitations
    - assumptions
    - generalization
    - external validity
14. Implementation validation
    - reference implementation
    - unit tests
    - randomized tests
    - sanity checks

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101. Resume & Interview Defense

Preparedness target: VERY STRONG

This is the final skill because ultimately **all the other skills exist to get you through placements**.

Subskills

1. Resume accuracy
    - every technology actually used
    - every metric defensible
    - no exaggerated claims
2. Project architecture explanation
    - components
    - data flow

- dependencies
- design decisions

3. Technical depth
    - why a technology was selected
    - alternatives
    - trade-offs
4. Implementation defense
    - what you personally implemented
    - difficult parts
    - debugging
    - optimization
5. Performance claims
    - benchmark methodology
    - baseline
    - workload
    - reproducibility
6. System-design defense
    - architecture
    - bottlenecks
    - scaling
    - failure modes
7. Database defense
    - schema
    - indexes
    - queries
    - transactions

8. Backend defense
    - API design
    - authentication
    - concurrency
    - deployment
9. ML-project defense
    - dataset
    - preprocessing
    - model
    - training
    - evaluation
10. Algorithm defense
    - algorithm choice
    - complexity
    - correctness
    - alternatives
11. Research defense
    - problem
    - prior work
    - contribution
    - limitations
12. "Why?" questioning
    - why this technology?
    - why this algorithm?
    - why this architecture?
    - why this metric?

13. "What if?" questioning
    - larger dataset
    - higher traffic
    - failure
    - adversarial input
    - insufficient resources
14. Trade-off questions
    - performance vs simplicity
    - consistency vs availability
    - memory vs speed
    - accuracy vs latency
15. Debugging questions
    - production failure
    - performance regression
    - incorrect output
    - race condition
16. Behavioral project questions
    - biggest challenge
    - mistake
    - conflict
    - learning
    - ownership
17. Project-specific interview simulation
    - easy questions
    - medium questions
    - deep-dive questions

- interviewer follow-ups

18. Resume-specific quiz generation
    - select resume
    - identify claimed technologies
    - map to subskills
    - generate questions
19. Interview readiness assessment
    - knowledge
    - confidence
    - demonstrated performance
    - weak areas
20. Overclaim detection
    - technology listed but poorly understood
    - suspiciously shallow knowledge
    - unsupported metrics
21. Final resume-defensibility criterion
    - "Can I explain this to a skeptical interviewer for 10 minutes?"